

# Xiaozhu Zhang

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## RESEARCH INTERESTS

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- Model selection, uncertainty quantification, multiple testing
- Causal discovery, domain adaptation

## EDUCATION

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**University of Washington, Seattle**

Ph.D. in **Statistics**, GPA: 3.94/4.00

Seattle, WA

09/2023 – 06/2028

**Duke University**

M.S. in **Statistical Science**, GPA: 4.00/4.00

Durham, NC

08/2021 – 05/2023

**Hunan University**

B.S. in **Statistics**, GPA: 3.94/4.00

Changsha, China

09/2016 – 07/2020

## RESEARCH EXPERIENCE

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**UW Department of Statistics | Advisor: Armeen Taeb**

Seattle, WA

*Research Assistant in **Causal Discovery***

03/2025 – Present

- Improved accuracy for nonlinear DAG recovery via convex mixed integer programming; outperformed all existing methods.
- Developed non-parametric theories including correct permutation, variance convergence rate, and perfect DAG recovery.

*Research Assistant in **Uncertainty Quantification** [software]*

10/2023 – 05/2025

- Developed an uncertainty-aware framework of model selection for nearly linearly dependent data; re-defined the notion of true positive, false positive error, stability, and substitutability via a subspace perspective.
- Designed an algorithm that returns models with false positive error control, high power and robustness; established a data analysis pipeline that calibrates and identifies models based on stability and substitutability.

**Duke Department of Statistical Science**

Durham, NC

*Research Assistant in **Stable Model Selection** [software] | Advisor: Yuansi Chen*

10/2021 – 05/2023

- Proved that Lasso solution set forms a polytope when multiple solution exist; conducted finite-sample theoretical analysis.
- Designed an MCMC algorithm that uniformly samples on this polytope of Lasso solutions.

*Research Assistant in **Non-standard Sampling** [software] | Advisor: Jerome Reiter*

07/2022 – 05/2023

- Developed a new sampling procedure for the generalized inverse Gaussian density; obtained the lowest rejection constants.

## PAPERS

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- **Zhang, X.**, Keret, N., Shojaie A., and Taeb, A. (2025). Convex Mixed-Integer Programming for Causal Additive Models with Optimization and Statistical Guarantees. *arXiv preprint arXiv:2511.21126*.
- **Zhang, X.**, Bien, J., and Taeb, A. (2025). Quantifying Uncertainty and Stability Among Highly Correlated Predictors: a Subspace Perspective. *arXiv preprint arXiv:2505.06760*.
- **Zhang, X.** (2023). Stable Variable Selection for Sparse Linear Regression in a Non-uniqueness Regime (**Master's thesis**, Duke University).
- **Zhang, X.**, and Reiter, J. P. (2022). A Generator for Generalized Inverse Gaussian Distributions. *arXiv preprint arXiv:2211.13049*.

## TALKS

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- The 5th Biennial Meeting of the Pacific Northwest Section of SIAM, October 2025, Seattle, WA
- The 3rd Joint Conference on Statistics and Data Science in China, July 2025, Hangzhou, China

## AWARDS

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- Dean's Research Award for Master's Students, Duke University, 2022
- Outstanding Undergraduates, Hunan Province, 2020
- China National Scholarship, 2017 – 2019
- National 1st Prize & Best Paper Award in China Undergraduate Mathematical Contest in Modeling, 2018